

Valerio Di Tommaso

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SUMMARY

Economics PhD candidate (Northwestern, expected 2026) across **forecasting** and time-series econometrics, **simulation of heterogeneous populations**, and **causal inference**. Research in **out-of-sample forecasting** of U.S. real activity from large financial-market datasets, and in **heterogeneous-agent models** of household spending, simulated at the scale of **one million households** in Python and MATLAB, to measure how much households spend out of **income windfalls**. Prior research roles at the European Central Bank and Bank of Italy.

EDUCATION

Ph.D., Economics , Northwestern University	Expected September 2026
<i>Fields: Applied Econometrics, Time Series Econometrics, Quantitative Macroeconomics</i>	
M.Sc., Economics , University of Turin	2018
M.A., Economics , Collegio Carlo Alberto	2018
B.A., Economics , Sapienza University of Rome	2016

SELECTED RESEARCH

Forecasting & Time Series

- Built predictors of market-implied uncertainty (volatility, skewness, tail risk) from **option-implied probability distributions (545,873 daily option observations**, OptionMetrics, 2008–2021) to forecast the **U.S. unemployment rate and industrial production** at the one-month horizon; in **expanding-window out-of-sample** tests this cut **forecast error by ~10%** (10.5% and 10.2% respectively) versus an AR benchmark with financial controls (VIX, oil price). ([Di Tommaso, 2026](#))

Population Simulation & Optimization

- Solved for the **optimal allocation of a fixed cash-transfer budget** across households to **maximize aggregate spending** under four information regimes: conditioning on **income alone captures only ~23%** of the gains over a uniform transfer, while **adding assets raises this to ~89%**, in a heterogeneous-household model of consumption and saving. ([Di Tommaso, 2026](#))
- Built and calibrated a **heterogeneous-agent consumption-saving model** with durable goods in **Python** (Numba-accelerated); solved the household **dynamic optimization** and simulated **one million households** to quantify how spending out of an income windfall varies with its size. ([Di Tommaso, 2025](#))

Causal Inference & Natural Experiments

- Showed that standard estimators (OLS, IV, Diff-in-Diff) of the **average treatment effect** of income windfalls on household spending in natural experiments are **biased downward**, overweighting larger windfalls when unit-level effects are heterogeneous; derived **partial-identification bounds** for the true average effect. ([Di Tommaso, 2025](#))
- On the **2008 U.S. tax rebate** (CEX data), a bias-corrected estimator increases the estimated average effect by **~43%** over standard estimates. ([Di Tommaso, 2025](#))
- Estimated **local projections** on a quarterly panel of **60 euro-area labour market reforms**; documented heterogeneous **wage and employment responses** across age groups and states of the economy. ([Aumond, Di Tommaso & Rünstler, ECB WP 2657, 2022](#))
- Characterized identifying assumptions for the **IV model with continuous treatments and instruments** under heterogeneous treatment effects; derived a **testable monotonicity implication** used as a robustness check across three empirical cases. ([Crippa & Di Tommaso, 2023](#))
- Built a new **instrumental variable** for oil supply shocks from daily shifts in the **option-implied distribution** around OPEC announcements, raising first-stage strength from an **effective F of 8.5 to 17**. ([Di Tommaso, 2026](#))

PROFESSIONAL EXPERIENCE

Research Assistant, European Central Bank (DG-Research) Sep 2019 – Aug 2020

- Maintained the **output and inflation gap models** underpinning the department’s policy initiatives, providing econometric support to cross-functional teams.
- Processed **high-frequency money-market transaction data (MMSR)** in **SQL** and Stata into a research-ready dataset for the Research Department.

Research Intern, Bank of Italy Jun 2021 – Aug 2021

- Built a monthly panel of **88 PPI and 93 HICP** euro-area price series (2000–2021); applied **factor models and PCA** to identify the series driving comovement, then quantified shock propagation with a **factor-augmented VAR**.

Graduate Research Assistant, Northwestern University (Prof. Diego Känzig) Jan 2023 – Jun 2023

- Built a dataset of U.S. **climate and EPA environmental regulation**, and developed a **structural multi-sector (DSGE) model** in MATLAB.

Teaching Assistant, Northwestern University Sep 2021 – Jun 2026

- Led review sessions for 20–100 students across **Macroeconomics, Microeconomics, Applied Econometrics, and Corporate Finance**, translating complex quantitative results for **technical and non-technical audiences**.

SKILLS

Programming & Tools: Python (Pandas, NumPy, SciPy, Scikit-learn, Numba, Statsmodels, Matplotlib), SQL, R, Stata, MATLAB, Excel, Claude Code.

Methods: Forecasting, Nowcasting & Time-Series Econometrics, Out-of-Sample Forecast Evaluation, Density Estimation, Structural & Heterogeneous-Agent Simulation, Optimization, Causal Inference, Quasi-Experimental Designs, A/B & Hypothesis Testing, Panel Data, Machine Learning (Dimensionality Reduction, Regularization, Tree-Based Methods), Data Cleaning & Visualization.

AWARDS

Northwestern Dissertation Year Fellowship (2025); Bank of Italy “Giorgio Mortara” Scholarship (2020); Collegio Carlo Alberto Honors Scholarship (2016).

LANGUAGES

English (fluent), Italian (native).